

1. Questions

Two equations are given below.

Equation I: $X^2 - AX + 72 = 0$

Equation II: $Y^2 - BY + 90 = 0$

The greater root of equation I is twice the smaller root of the same equation.

The difference between both roots of equation II is 13.

Which of the following is/are must be true?

I: Both [A] and [B] are odd numbers.

II: [A] is exactly divisible by 3.

III: The product of the digits of [B] is 6.

- a. Both I and III
- b. Both II and III
- c. Only II
- d. Only III
- e. All I, II and III

2. Questions

Two equations are given below.

Equation I: $MP^2 - 27P + K = 0$

Equation II: $NQ^2 - 34Q + L = 0$

M and N are consecutive single-digit natural numbers, which are exactly divisible by 3. ($M > N$)

Neither M nor N is a perfect square.

K and L are odd integers more than 20 but less than 33.

The difference between K and L is 10. ($K < L$)

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Find the sum of the greater root of equation I and the smaller root of equation II.

- a. 3.50
- b. 2.25
- c. 6.75
- d. 8.25
- e. 4.50

3. Questions

Three equations are given below.

Equation I: $A^2 - A - X = 0$

Equation II: $B^2 - 12B + Y = 0$

Equation III: $C^2 + 5C - Z = 0$

The product of the positive roots of equations I and III is 12 and their sum is a perfect cube. ($I > III$)

The difference between the negative roots of Equation I and III is equal to the smaller root of Equation II.

Determine the correct relation between [X], [Y], and [Z].

- a. $X < Y < Z$
- b. $X > Y > Z$
- c. $X > Y < Z$
- d. $X < Y > Z$
- e. $X = Y = Z$

4. Questions

An equation is given below.

Equation: $PK^2 - QK + R = 0$

P is a single-digit prime number and R is a two-digit prime number.

The sum of P and R is equal to Q.

The sum of the digits of R is 5.

The square of R is less than 1000.

Q is exactly divisible by 3.

Which of the following is/are definitely true?

I: $P^2 = 49$

II: $\sqrt{Q - 5} = 5$

III: $(R/23) = 1$

- a. Both II and III
- b. Only I
- c. Both I and II
- d. All I, II and III

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e. Only III

5. Questions

An equation is given below.

Equation: $M^2 + (N^3) M - L = 0$

N is an odd integer not more than 5.

The positive root of the equation is 7.

L lies somewhere between 200 and 300.

Find the roots of the equation given above.

- a. 5 and – 25
- b. 9 and – 13
- c. 7 and – 34
- d. 3 and – 41
- e. 1 and – 50

6. Questions

Read the following information carefully and answer the questions based on it.

Three bags P, Q, and R contain four different colours of balls namely blue, green, red, and yellow.

Bag P: The number of blue balls is 20 more than the number of yellow balls in the bag. The number of green balls is 10 less than the number of blue balls in the bag. The number of red balls is 10 more than the number of yellow balls in the bag. The probability of picking one yellow ball from the bag is $\frac{1}{8}$.

Bag Q: The number of blue balls in the bag is 20 more than the number of red balls in the same bag. The number of green balls in the bag is 3 times the number of yellow balls in the same bag. The number of yellow balls in bag P and Q are the same. The probability of picking one blue ball from the bag is $\frac{2}{5}$.

Bag R: The number of red balls in the bag is 10, which is 33.33% of the number of blue balls in the same bag. The number of green balls in the bag is 10 more than the number of yellow balls in the same bag. The probability of picking one green ball from the bag is $\frac{1}{4}$.

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Note:

The total number of balls in the bag = the number of blue balls + the number of green balls + the number of red balls + the number of yellow balls

The probability of picking one red ball from bag P is what percent of the probability of picking one blue ball from the same bag?

- a. 71.42%
- b. 83.33%

- c. 37.50%
- d. 66.67%
- e. 62.50%

7. Questions

If the number of pink balls in bag Q is [A], then the probability of picking one blue ball from the bag Q will be $[8/A]$. Find the ratio of the number of pink and yellow balls in bag Q. (The number of green, blue, red and yellow balls remained unchanged)

- a. 5:2
- b. 3:1
- c. 2:1
- d. 4:2
- e. 7:2

8. Questions

In another bag S, the number of blue balls is $[2X]$, which is $2/3^{\text{rd}}$ of the number of green balls and the number of yellow balls is $[3Y]$, which is $3/7^{\text{th}}$ of the number of red balls. If the total number of balls in bag S is 50 and the probability of picking one blue ball from bag S is $[Y + 4]/25$, then find the number of green balls in bag S.

- a. 21
- b. 18
- c. 15
- d. 24
- e. 9

9. Questions

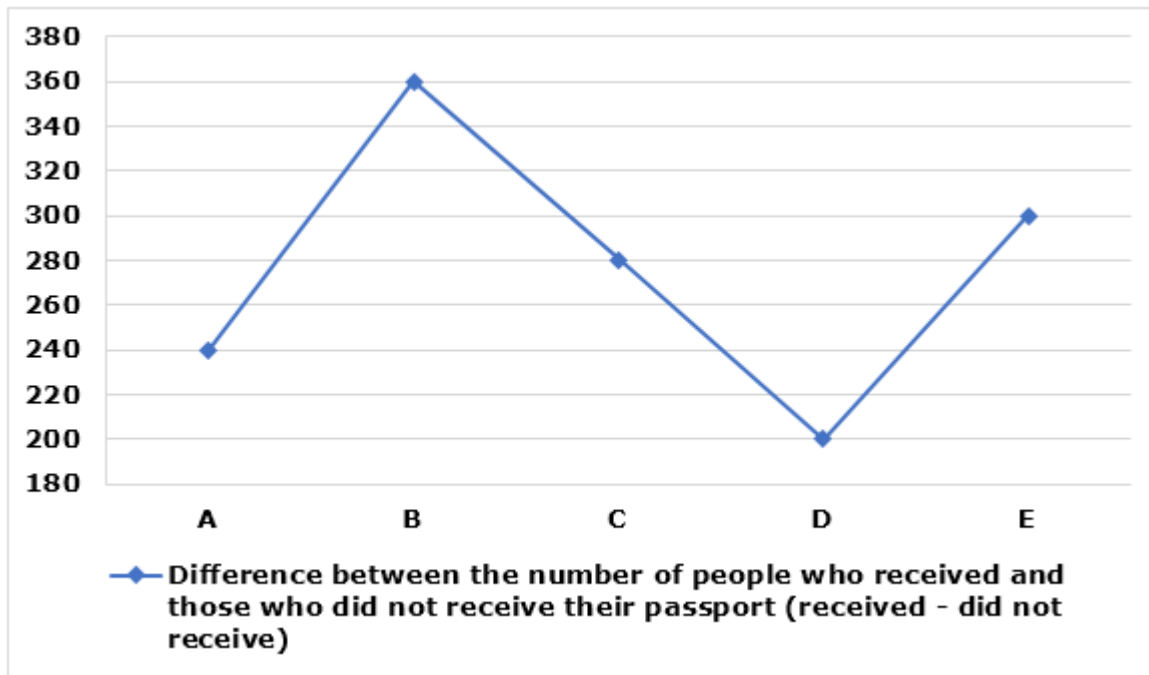
Find the sum of the number of blue balls in bag Q and the number of yellow balls in bag R.

- a. 24
- b. 48
- c. 45
- d. 55
- e. 60

10. Questions

Read the following information carefully and answer the questions based on it.

The line graph given below shows the difference between the number of people who received and those who did not receive their passport (received – did not receive) in five different cities A, B, C, D and E.



Notes:

- i). The total number of people who applied for the passport = the number of people whose applications were accepted + the number of people whose applications were rejected
- ii). The total number of people whose applications were accepted = number of people who received their passports + the number of people who did not receive their passports.
- iii). In each city, the number of people who received their passports is more than the number of people who did not receive their passports.
- iv). The total number of people who applied for the passport in cities A, B, C, D, and E is 400, 800, 600, 500 and 900 respectively.
- v). The ratio of the number of people who received their passports in cities A and B is 7:12.
- vi). The difference between the number of people whose applications were rejected in cities A and B is 120. ($A < B$)
- vii). The total number of people who received their passports in cities D and E together is 875.
- viii). The number of people whose application got rejected in city D is 50% of city E.
- ix). In city C, the number of people who did not receive their passports is 110 less than the number of people whose applications were rejected in the same city.

The number of people who did not receive their passport in city B is what percent of the total number of people whose application got accepted in the same city?

- a. 20%

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- b. 30%
- c. 40%
- d. 50%
- e. 60%

11. Questions

In city D, if the number of females whose applications were rejected is 10, which is 10% of the number of females whose applications were accepted, then find the total number of males who applied for the passport from city D.

- a. 420
- b. 280
- c. 500
- d. 160
- e. 390

12. Questions

In city C, the total number of people who applied for the driving license is 500 and the number of people who received driving license is 50 less than the number of people who received passport. If the number of people who did not receive driving license is equal to the number of people whose applications for driving license were rejected, then find total number of people whose applications were accepted for driving license in city C.

- a. 400
- b. 200
- c. 240
- d. 360
- e. 180

13. Questions

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Find the average number of people whose applications for passport were rejected in all five cities together.

- a. 156
- b. 108
- c. 172
- d. 122
- e. 190

14. Questions

If the total number of people who applied for the passport in another city F is 60% of city E and the number of people whose applications were rejected in city F is $\frac{1}{9}$ of the total number of people who applied from the same city, then find the number of people whose applications were accepted in city F.

- a. 420
- b. 560
- c. 480
- d. 380
- e. 240

15. Questions

Find the ratio of the number of people who received their passport in city A and that in city D, respectively.

- a. 65:56
- b. 56:65
- c. 42:37
- d. 21:29
- e. 33:31

16. Questions

The following question has two series as series I and series II. Find the pattern of series I and series II in each question and give answer accordingly.

The series I has two missing terms as [A] and [B] and the series II has two missing terms as [P] and [Q]. Both series follow same pattern.

Series I: 10, 14, -2, [A], -30, [B]

Series II: 15, [P], 3, [Q], -25, 75

Which of the following is/are must be true?

I: [Q] is negative integer.

II: The difference between [P] and [A] is 19.

III: The product of [B] and [Q] is four-digit natural number.

- a. Only III
- b. Both I and III
- c. Both II and III

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- d. Only II
- e. All I, II and III

17. Questions

The series I has one wrong term and the series II starts with correct term that will replace the wrong term of series I and the both series follows different pattern.

Series I: 24, 48, 12, 30, 6, 12

Series II: ____, 51, 115, 240, 456, __[?]

Find the value of question mark.

- a. 812
- b. 666
- c. 901
- d. 350
- e. 799

18. Questions

The series I and II have one wrong term each and the both series follow same pattern.

Series I: 12, 23, 36, 53, 70, 95

Series II: 31, 42, 55, 72, 91, 110

$[\sqrt{M} + 2]$ = The difference between the digits of the correct term that will replace the wrong term of series I.

$[N^2]$ = The square of the unit digit of the correct term that will replace the wrong term of series II.

- a. $M > N$
- b. $M < N$
- c. $M \geq N$
- d. $M \leq N$
- e. $M = N$

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19. Questions

Series I and II has one missing term each as [X] and [Y] respectively. Both series follow different pattern.

Series I: 8, 21, 58, 237, __[X]__, 7085

Series II: $[X+30/110]$, 19, 35, 59, 91, __[Y]__

Find the value of $[10Y - X]$.

- a. 120
- b. 245
- c. 75
- d. 100
- e. 130

20. Questions

The series I has missing term as [m] and the series II has one wrong term. The both series follow same pattern.

Series I: 10, 22, 69, 280, __[m]__, 8436

Series II: [(m - 5)/280], 12, 39, 156, 805, 4836

Find the correct term that will replace the wrong term of series II.

- a. 43
- b. 160
- c. 15
- d. 4830
- e. 816

Explanations:

1. Questions

Answer: B

Equation I

The ratio of greater and smaller roots = 2:1

The sum of the roots = A

The greater root = $\frac{2}{3}$ of A = $\frac{2A}{3}$

The smaller root = $\frac{1}{3}$ of A = $\frac{A}{3}$

The product of the roots = 72

Now,

$$\frac{2A}{3} * \frac{A}{3} = 72$$

$$2A^2 = 72 * 9$$

$$2A^2 = 648$$

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$$A^2 = 324$$

$$A = \sqrt{324}$$

$$A = 18$$

Equation II

The sum of the roots = B

The greater root = $(B + 13)/2$

The smaller root = $(B - 13)/2$

The product of the roots = 90

Now,

$$(B + 13)/2 * (B - 13)/2 = 90$$

$$(B^2 - 13B + 13B - 169)/4 = 90$$

$$B^2 - 13B + 13B - 169 = 360$$

$$B^2 = 360 + 169$$

$$B^2 = 529$$

$$B = \sqrt{529}$$

$$B = 23$$

I: 18 is not an odd number.

So, I is not true.

$$\text{II: } 18/3 = 6$$

A is exactly divisible by 3.

So, II is true.

$$\text{III: The product of the digits of } 23 = (2 * 3) = 6$$

So, III is true.

Therefore, both II and III must be true.

Hence, the correct answer is option B.

2. Questions

Answer: E

The only possible values of M and N = 3 and 6

So, M = 6

And, N = 3

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The only possible values of K and L = 21 and 31

So, $K = 21$

And, $L = 31$

Equation I: $MP^2 - 27P + K = 0$

$$6P^2 - 27P + 21 = 0$$

$$6P^2 - 21P - 6P + 21 = 0$$

$$3P(2P - 7) - 3(2P - 7) = 0$$

$$(3P - 3)(2P - 7) = 0$$

$$P = +1 \text{ or } +3.50$$

So, the greater root = 3.50

Equation II: $NQ^2 - 34Q + L = 0$

$$3Q^2 - 34Q + 31 = 0$$

$$3Q^2 - 31Q - 3Q + 31 = 0$$

$$Q(3Q - 31) - 1(3Q - 31) = 0$$

$$(Q - 1)(3Q - 31) = 0$$

$$Q = +1 \text{ or } +10.33$$

So, the smaller root = 1

Therefore, the required sum = $(3.50 + 1) = 4.50$.

Hence, the correct answer is option E.

3. Questions

Answer: B

The product of the positive roots of equation I and III = 12

The factors of 12 = $(1 * 2 * 2 * 3 * 4 * 6 * 12)$

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So, the possible value of the positive root of equation I = 6 or 4 or 12

And, the possible value of the positive root of equation III = 2 or 3 or 1

The sum of the positive roots of equation I and III = $(6 + 2) = 8$ (perfect cube)

Also, the sum of the positive roots of equations I and III = $(4 + 3) = 7$ (not perfect cube)

Also, the sum of the positive roots of equations I and III = $(12 + 1) = 13$ (not perfect cube)

So, the positive root of equation I = 6

And, the positive root of equation III = 2

Equation I: $A^2 - A - X = 0$

The sum of the roots = 1

The positive root = 6

So, the negative root = $(1 - 6) = -5$

Now, the product of the roots = $(6 * -5) = -30$

So, $X = 30$

Equation III: $C^2 + 5C - Z = 0$

The sum of the roots = -5

The positive root = 2

So, the negative root = $(-5 - 2) = -7$

The product of the roots = $(2 * -7) = -14$

So, $Z = 14$

Equation II: $B^2 - 12B + Y = 0$

The sum of the roots = 12

The smaller root = $-5 - (-7) = (-5 + 7) = 2$

So, the greater root = $(12 - 2) = 10$

The product of the roots = $(2 * 10) = 20$

So, $Y = 20$

Now,

$X = 30$

$Y = 20$

$Z = 14$

Therefore, $X > Y > Z$.

Hence, the correct answer is option B.

4. Questions

Answer: D

The possible values of $R = 41$ or 23

The square of $41 = 41^2 = 1681$ (not possible)

The square of $23 = 23^2 = 529$ (possible)

So, $R = 23$

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The possible values of $P = 2, 3, 5$ or 7

So,

$$(23 + 2) = 25 \text{ (not possible)}$$

$$(23 + 3) = 26 \text{ (not possible)}$$

$$(23 + 5) = 28 \text{ (not possible)}$$

$$(23 + 7) = 30 \text{ (possible)}$$

$$\text{So, } P = 7$$

Also,

$$(P + R) = (7 + 23) = 30$$

$$\text{So, } Q = 30 \text{ (exactly divisible by 3)}$$

$$\text{I: } P^2 = 7^2 = 49$$

So, I is definitely true.

$$\text{II: } \sqrt{(Q - 5)} = \sqrt{(30 - 5)} = \sqrt{25} = 5$$

So, II is definitely true.

$$\text{III: } (R/23) = (23/23) = 1$$

So, III is definitely true.

Therefore, all I, II, and III are definitely true.

Hence, the correct answer is option D.

5. Questions

Answer: C

The possible values of $N = 1, 3$ or 5

Case I

$$\text{Let } N = 1$$

$$\text{So, the sum of the roots} = -1^3 = -1$$

$$\text{The positive root} = 7$$

$$\text{So, the negative root} = (-1 - 7) = -8$$

$$\text{The product of the roots} = (-8 * 7) = -56$$

$$L = 56 \text{ (not possible)}$$

Case II

$$\text{Let } N = 3$$

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So, the sum of the roots = $-3^3 = -27$

The positive root = 7

So, the negative root = $(-27 - 7) = -34$

The product of the roots = $(-34 * 7) = -238$

L = 238 (possible)

Case III

Let N = 5

So, the sum of the roots = $-5^3 = -125$

The positive root = 7

So, the negative root = $(-125 - 7) = -132$

The product of the roots = $(-132 * 7) = -924$

L = 924 (not possible)

Now,

N = 3

L = 238

Equation: $M^2 + (N^3) M - L = 0$

$$M^2 + (3^3) M - 238 = 0$$

$$M^2 + 27M - 238 = 0$$

$$M^2 + 34M - 7M - 238 = 0$$

$$M(M + 34) - 7(M + 34) = 0$$

$$(M - 7)(M + 34) = 0$$

$$M = +7 \text{ or } -34$$

So, the roots are 7 and -34.

Hence, the correct answer is option C.

6. Questions

Bag P

Let the number of yellow balls be k.

So, the number of blue balls = $(k + 20)$

The number of green balls = $(k + 20) - 10 = (k + 10)$

The number of red balls = $(k + 10)$

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The total number of balls = $k + (k + 20) + (k + 10) + (k + 10) = (4k + 40)$

So, the probability of picking one yellow ball = $k / (4k + 40)$

Now,

$$k / (4k + 40) = 1/8$$

$$8k = 4k + 40$$

$$4k = 40$$

$$k = 10$$

Therefore,

The number of yellow balls = $k = 10$

The number of blue balls = $(k + 20) = (10 + 20) = 30$

The number of green balls = $(k + 10) = (10 + 10) = 20$

The number of red balls = $(k + 10) = (10 + 10) = 20$

The total number of balls = $(10 + 30 + 20 + 20) = 80$

Bag Q

Let the number of red balls be m .

So, the number of blue balls = $(m + 20)$

The number of yellow balls = 10

The number of green balls = $(10 * 3) = 30$

The total number of balls = $m + (m + 20) + 10 + 30 = (2m + 60)$

The probability of picking one blue ball = $(m + 20) / (2m + 60)$

Now,

$$(m + 20) / (2m + 60) = 2/5$$

$$5m + 100 = 4m + 120$$

$$m = 20$$

Therefore,

The number of red balls = $m = 20$

The number of blue balls = $(m + 20) = (20 + 20) = 40$

The total number of balls = $(20 + 40 + 10 + 30) = 100$

Bag R

The number of red balls = 10

The number of blue balls = $10/1 * 3 = 30$ (**Note: 33.33% = 1/3**)

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Let the number of green balls be n .

The number of yellow balls = $(n - 10)$

The total number of balls = $10 + 30 + n + (n - 10) = (2n + 30)$

The probability of picking green red ball = $n / (2n + 30)$

Now,

$$n / (2n + 30) = 1/4$$

$$4n = 2n + 30$$

$$2n = 30$$

$$n = 15$$

Therefore,

The number of green balls = $n = 15$

The number of yellow balls = $(n - 10) = (15 - 10) = 5$

The total number of balls = $(10 + 30 + 15 + 5) = 60$

Bag	The number of blue balls	The number of green balls	The number of red balls	The number of yellow balls	The total number of balls
P	30	20	20	10	80
Q	40	30	20	10	100
R	30	15	10	5	60
Total	100	65	50	25	240

Answer: D

Now, according to the question,

Bag P

So, the probability of picking one red ball = $20/80 = 1/4$

And, the probability of picking one blue ball = $30/80 = 3/8$

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Therefore, the required percentage = $[(1/4) / (3/8)] * 100 = 66.67\%$.

Hence, the correct answer is option D.

7. Questions

Bag P

Let the number of yellow balls be k .

So, the number of blue balls = $(k + 20)$

The number of green balls = $(k + 20) - 10 = (k + 10)$

The number of red balls = $(k + 10)$

The total number of balls = $k + (k + 20) + (k + 10) + (k + 10) = (4k + 40)$

So, the probability of picking one yellow ball = $k / (4k + 40)$

Now,

$$k / (4k + 40) = 1/8$$

$$8k = 4k + 40$$

$$4k = 40$$

$$k = 10$$

Therefore,

The number of yellow balls = $k = 10$

The number of blue balls = $(k + 20) = (10 + 20) = 30$

The number of green balls = $(k + 10) = (10 + 10) = 20$

The number of red balls = $(k + 10) = (10 + 10) = 20$

The total number of balls = $(10 + 30 + 20 + 20) = 80$

Bag Q

Let the number of red balls be m .

So, the number of blue balls = $(m + 20)$

The number of yellow balls = 10

The number of green balls = $(10 * 3) = 30$

The total number of balls = $m + (m + 20) + 10 + 30 = (2m + 60)$

The probability of picking one blue ball = $(m + 20) / (2m + 60)$

Now,

$$(m + 20) / (2m + 60) = 2/5$$

$$5m + 100 = 4m + 120$$

$$m = 20$$

Therefore,

The number of red balls = $m = 20$

The number of blue balls = $(m + 20) = (20 + 20) = 40$

The total number of balls = $(20 + 40 + 10 + 30) = 100$

Bag R

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The number of red balls = 10

The number of blue balls = $10/1 \times 3 = 30$ (*Note: 33.33% = 1/3*)

Let the number of green balls be n.

The number of yellow balls = $(n - 10)$

The total number of balls = $10 + 30 + n + (n - 10) = (2n + 30)$

The probability of picking green red ball = $n / (2n + 30)$

Now,

$$n / (2n + 30) = 1/4$$

$$4n = 2n + 30$$

$$2n = 30$$

$$n = 15$$

Therefore,

The number of green balls = $n = 15$

The number of yellow balls = $(n - 10) = (15 - 10) = 5$

The total number of balls = $(10 + 30 + 15 + 5) = 60$

Bag	The number of blue balls	The number of green balls	The number of red balls	The number of yellow balls	The total number of balls
P	30	20	20	10	80
Q	40	30	20	10	100
R	30	15	10	5	60
Total	100	65	50	25	240

Answer: A

Bag Q

The number of pink balls = A

And, the total number of green, blue, red and yellow balls = 100

Now, the total number of balls = $(100 + A)$

The number of blue balls = 40

So, the probability of picking one blue ball = $40 / (100 + A)$

Now,

$$40 / (100 + A) = 8 / A$$

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$$40A = 800 + 8A$$

$$32A = 800$$

$$A = 25$$

So, the number of pink balls = $A = 25$

And, the number of yellow balls = 10

Therefore, the required ratio = $25:10 = 5:2$.

Hence, the correct answer is option A.

8. Questions

Bag P

Let the number of yellow balls be k .

So, the number of blue balls = $(k + 20)$

The number of green balls = $(k + 20) - 10 = (k + 10)$

The number of red balls = $(k + 10)$

The total number of balls = $k + (k + 20) + (k + 10) + (k + 10) = (4k + 40)$

So, the probability of picking one yellow ball = $k / (4k + 40)$

Now,

$$k / (4k + 40) = 1/8$$

$$8k = 4k + 40$$

$$4k = 40$$

$$k = 10$$

Therefore,

The number of yellow balls = $k = 10$

The number of blue balls = $(k + 20) = (10 + 20) = 30$

The number of green balls = $(k + 10) = (10 + 10) = 20$

The number of red balls = $(k + 10) = (10 + 10) = 20$

The total number of balls = $(10 + 30 + 20 + 20) = 80$

Bag Q

Let the number of red balls be m .

So, the number of blue balls = $(m + 20)$

The number of yellow balls = 10

The number of green balls = $(10 * 3) = 30$

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The total number of balls = $m + (m + 20) + 10 + 30 = (2m + 60)$

The probability of picking one blue ball = $(m + 20) / (2m + 60)$

Now,

$$(m + 20) / (2m + 60) = 2/5$$

$$5m + 100 = 4m + 120$$

$$m = 20$$

Therefore,

The number of red balls = $m = 20$

The number of blue balls = $(m + 20) = (20 + 20) = 40$

The total number of balls = $(20 + 40 + 10 + 30) = 100$

Bag R

The number of red balls = 10

The number of blue balls = $10/1 \times 3 = 30$ (*Note: 33.33% = 1/3*)

Let the number of green balls be n .

The number of yellow balls = $(n - 10)$

The total number of balls = $10 + 30 + n + (n - 10) = (2n + 30)$

The probability of picking green red ball = $n / (2n + 30)$

Now,

$$n / (2n + 30) = 1/4$$

$$4n = 2n + 30$$

$$2n = 30$$

$$n = 15$$

Therefore,

The number of green balls = $n = 15$

The number of yellow balls = $(n - 10) = (15 - 10) = 5$

The total number of balls = $(10 + 30 + 15 + 5) = 60$

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Bag	The number of blue balls	The number of green balls	The number of red balls	The number of yellow balls	The total number of balls
P	30	20	20	10	80
Q	40	30	20	10	100
R	30	15	10	5	60
Total	100	65	50	25	240

Answer: B

Bag S

The number of,

Blue balls = $2X$

Green balls = $2X/2 \times 3 = 3X$

Yellow balls = $3Y$

Red balls = $3Y/3 \times 7 = 7Y$

Now,

$$2X + 3X + 3Y + 7Y = 50$$

$$5X + 10Y = 50$$

$$X + 2Y = 10 \dots\dots\dots(i)$$

Also, the probability of picking one blue ball = $2X / 50$

Now,

$$2X / 50 = (Y + 4) / 25$$

$$50X = 50Y + 200$$

$$50X - 50Y = 200$$

$$X - Y = 4 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$3Y = 6$$

$$Y = 2$$

Now, put the value of Y in equation (i), we get,

$$X + 2Y = 10$$

$$X + (2 \times 2) = 10$$

$$X = 10 - 4$$

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$$X = 6$$

Therefore, the number of green balls in bag S = $3X = (3 * 6) = 18$.

Hence, the correct answer is option B.

9. Questions

Bag P

Let the number of yellow balls be k .

So, the number of blue balls = $(k + 20)$

The number of green balls = $(k + 20) - 10 = (k + 10)$

The number of red balls = $(k + 10)$

The total number of balls = $k + (k + 20) + (k + 10) + (k + 10) = (4k + 40)$

So, the probability of picking one yellow ball = $k / (4k + 40)$

Now,

$$k / (4k + 40) = 1/8$$

$$8k = 4k + 40$$

$$4k = 40$$

$$k = 10$$

Therefore,

The number of yellow balls = $k = 10$

The number of blue balls = $(k + 20) = (10 + 20) = 30$

The number of green balls = $(k + 10) = (10 + 10) = 20$

The number of red balls = $(k + 10) = (10 + 10) = 20$

The total number of balls = $(10 + 30 + 20 + 20) = 80$

Bag Q

Let the number of red balls be m .

So, the number of blue balls = $(m + 20)$

The number of yellow balls = 10

The number of green balls = $(10 * 3) = 30$

The total number of balls = $m + (m + 20) + 10 + 30 = (2m + 60)$

The probability of picking one blue ball = $(m + 20) / (2m + 60)$

Now,

$$(m + 20) / (2m + 60) = 2/5$$

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$$5m + 100 = 4m + 120$$

$$m = 20$$

Therefore,

$$\text{The number of red balls} = m = 20$$

$$\text{The number of blue balls} = (m + 20) = (20 + 20) = 40$$

$$\text{The total number of balls} = (20 + 40 + 10 + 30) = 100$$

Bag R

$$\text{The number of red balls} = 10$$

$$\text{The number of blue balls} = 10/1 \times 3 = 30 \text{ (Note: } 33.33\% = 1/3)$$

Let the number of green balls be n .

$$\text{The number of yellow balls} = (n - 10)$$

$$\text{The total number of balls} = 10 + 30 + n + (n - 10) = (2n + 30)$$

$$\text{The probability of picking green red ball} = n / (2n + 30)$$

Now,

$$n / (2n + 30) = 1/4$$

$$4n = 2n + 30$$

$$2n = 30$$

$$n = 15$$

Therefore,

$$\text{The number of green balls} = n = 15$$

$$\text{The number of yellow balls} = (n - 10) = (15 - 10) = 5$$

$$\text{The total number of balls} = (10 + 30 + 15 + 5) = 60$$

Bag	The number of blue balls	The number of green balls	The number of red balls	The number of yellow balls	The total number of balls
P	30	20	20	10	80
Q	40	30	20	10	100
R	30	15	10	5	60
Total	100	65	50	25	240

Answer: C

Bag Q

The number of blue balls in bag Q = 40

The number of yellow balls in bag R = 5

Therefore, the required sum = $(40 + 5) = 45$.

Hence, the correct answer is option C.

10. Questions

City A and B

Let the number of people who received their passports in cities A and B be $7m$ and $12m$ respectively.

So, the number of people who did not receive their passports in city A = $(7m - 240)$

And, the number of people who did not receive their passports in city B = $(12m - 360)$

So, the total number of people whose applications were accepted in city A = $7m + (7m - 240) = (14m - 240)$

And, the total number of people whose applications were accepted in city B = $12m + (12m - 360) = (24m - 360)$

So, the number of people whose applications were rejected in city A = $400 - (14m - 240) = (640 - 14m)$

And, the number of people whose applications were rejected in city B = $800 - (24m - 360) = (1160 - 24m)$

Now,

$$(1160 - 24m) - (640 - 14m) = 120$$

$$520 - 10m = 120$$

$$10m = 400$$

$$m = 40$$

City D and E

Let the number of people who received their passport in city D be n .

So, the number of people who received their passport in city E = $(875 - n)$

The number of people who did not receive their passport in city D = $(n - 200)$

And, the number of people who did not receive their passport in city E = $(875 - n) - 300 = (575 - n)$

So, the number of people whose applications were accepted in city D = $n + (n - 200) = (2n - 200)$

And, the number of people whose applications were accepted in city E = $(875 - n) + (575 - n) = (1450 - 2n)$

Now, the number of people whose applications were rejected in city D = $500 - (2n - 200) = (700 - 2n)$

And, the number of people whose applications were rejected in city E = $900 - (1450 - 2n) = (2n - 550)$

So,

$$(700 - 2n) / (2n - 550) = 50/100$$

$$(700 - 2n) / (2n - 550) = 1/2$$

$$1400 - 4n = 2n - 550$$

$$6n = 1950$$

$$n = 325$$

City A

The total number of people who applied = 400

The number of people whose application were accepted = $(14m - 240) = (14 * 40 - 240) = (560 - 240) = 320$

The number of people whose application were rejected = $(400 - 320) = 80$

The number of people who received their passports = $7m = (7 * 40) = 280$

The number of people who did not receive their passports = $(280 - 240) = 40$

City B

The total number of people who applied = 800

The number of people whose applications were accepted = $(24m - 360) = (24 * 40 - 360) = (960 - 360) = 600$

The number of people whose applications were rejected = $(800 - 600) = 200$

The number of people who received their passport = $12m = (12 * 40) = 480$

The number of people who did not receive their passport = $(480 - 360) = 120$

City D

The total number of people who applied = 500

The number of people whose applications were accepted = $(2n - 200) = (2 * 325 - 200) = (650 - 200) = 450$

The number of people whose applications were rejected = $(500 - 450) = 50$

The number of people who received their passport = $n = 325$

The number of people who did not receive their passport = $(325 - 200) = 125$

City E

The total number of people who applied = 900

The number of people whose applications were accepted = $(1450 - 2n) = (1450 - 2 * 325) = (1450 - 650) = 800$

The number of people whose applications were rejected = $(900 - 800) = 100$

The number of people who receive their passport = $(875 - n) = (875 - 325) = 550$

The number of people who did not receive their passport = $(550 - 300) = 250$

City C

The total number of people who applied = 600

Let the number of people whose applications were rejected be k .

So, the number of people whose applications were accepted = $(600 - k)$

The number of people who did not receive their passport = $(k - 110)$

The number of people who received their passport = $(k - 110) + 280 = (k + 170)$

Now,

$$(k - 110) + (k + 170) = (600 - k)$$

$$2k + 60 = 600 - k$$

$$3k = 540$$

$$k = 180$$

Therefore,

The number of people whose applications were rejected = $k = 180$

The number of people whose applications were accepted = $(600 - k) = (600 - 180) = 420$

The number of people who did not receive their passport = $(k - 110) = (180 - 110) = 70$

The number of people who received their passport = $(k + 170) = (180 + 170) = 350$

City	The total number of people who applied	The number of people whose application were accepted	The number of people whose application were rejected	The number of people who received their passport	The number of people who did not receive their passport
A	400	320	80	280	40
B	800	600	200	480	120
C	600	420	180	350	70
D	500	450	50	325	125
E	900	800	100	550	250
Total	3200	2590	610	1985	605

Answer: A

Now, according to the question,

City B

The number of people who did not receive their passport = 120

The total number of people whose application got accepted = 600

Therefore, the required percentage = $120/600 \times 100 = 20\%$.

Hence, the correct answer is option A.

11. Questions

City A and B

Let the number of people who received their passports in cities A and B be $7m$ and $12m$ respectively.

So, the number of people who did not receive their passports in city A = $(7m - 240)$

And, the number of people who did not receive their passports in city B = $(12m - 360)$

So, the total number of people whose applications were accepted in city A = $7m + (7m - 240) = (14m - 240)$

And, the total number of people whose applications were accepted in city B = $12m + (12m - 360) = (24m - 360)$

So, the number of people whose applications were rejected in city A = $400 - (14m - 240) = (640 - 14m)$

And, the number of people whose applications were rejected in city B = $800 - (24m - 360) = (1160 - 24m)$

Now,

$$(1160 - 24m) - (640 - 14m) = 120$$

$$520 - 10m = 120$$

$$10m = 400$$

$$m = 40$$

City D and E

Let the number of people who received their passport in city D be n .

So, the number of people who received their passport in city E = $(875 - n)$

The number of people who did not receive their passport in city D = $(n - 200)$

And, the number of people who did not receive their passport in city E = $(875 - n) - 300 = (575 - n)$

So, the number of people whose applications were accepted in city D = $n + (n - 200) = (2n - 200)$

And, the number of people whose applications were accepted in city E = $(875 - n) + (575 - n) = (1450 - 2n)$

Now, the number of people whose applications were rejected in city D = $500 - (2n - 200) = (700 - 2n)$

And, the number of people whose applications were rejected in city E = $900 - (1450 - 2n) = (2n - 550)$

So,

$$(700 - 2n) / (2n - 550) = 50/100$$

$$(700 - 2n) / (2n - 550) = 1/2$$

$$1400 - 4n = 2n - 550$$

$$6n = 1950$$

$$n = 325$$

City A

The total number of people who applied = 400

The number of people whose application were accepted = $(14m - 240) = (14 * 40 - 240) = (560 - 240) = 320$

The number of people whose application were rejected = $(400 - 320) = 80$

The number of people who received their passports = $7m = (7 * 40) = 280$

The number of people who did not receive their passports = $(280 - 240) = 40$

City B

The total number of people who applied = 800

The number of people whose applications were accepted = $(24m - 360) = (24 * 40 - 360) = (960 - 360) = 600$

The number of people whose applications were rejected = $(800 - 600) = 200$

The number of people who received their passport = $12m = (12 * 40) = 480$

The number of people who did not receive their passport = $(480 - 360) = 120$

City D

The total number of people who applied = 500

The number of people whose applications were accepted = $(2n - 200) = (2 * 325 - 200) = (650 - 200) = 450$

The number of people whose applications were rejected = $(500 - 450) = 50$

The number of people who received their passport = $n = 325$

The number of people who did not receive their passport = $(325 - 200) = 125$

City E

The total number of people who applied = 900

The number of people whose applications were accepted = $(1450 - 2n) = (1450 - 2 * 325) = (1450 - 650) = 800$

The number of people whose applications were rejected = $(900 - 800) = 100$

The number of people who receive their passport = $(875 - n) = (875 - 325) = 550$

The number of people who did not receive their passport = $(550 - 300) = 250$

City C

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The total number of people who applied = 600

Let the number of people whose applications were rejected be k .

So, the number of people whose applications were accepted = $(600 - k)$

The number of people who did not receive their passport = $(k - 110)$

The number of people who received their passport = $(k - 110) + 280 = (k + 170)$

Now,

$$(k - 110) + (k + 170) = (600 - k)$$

$$2k + 60 = 600 - k$$

$$3k = 540$$

$$k = 180$$

Therefore,

The number of people whose applications were rejected = $k = 180$

The number of people whose applications were accepted = $(600 - k) = (600 - 180) = 420$

The number of people who did not receive their passport = $(k - 110) = (180 - 110) = 70$

The number of people who received their passport = $(k + 170) = (180 + 170) = 350$

City	The total number of people who applied	The number of people whose application were accepted	The number of people whose application were rejected	The number of people who received their passport	The number of people who did not receive their passport
A	400	320	80	280	40
B	800	600	200	480	120
C	600	420	180	350	70
D	500	450	50	325	125
E	900	800	100	550	250
Total	3200	2590	610	1985	605

Answer: E

City D

The total number of people whose applications were rejected = 50

The number of females whose applications were rejected = 10

The number of males whose applications were rejected = $(50 - 10) = 40$

The total number of people whose applications were accepted = 450

The number of females whose applications were accepted = $10/10 \times 100 = 100$

The number of males whose applications were accepted = $(450 - 100) = 350$

Therefore, the total number males who applied for the passport = $(350 + 40) = 390$.

Hence, the correct answer is option E.

12. Questions

City A and B

Let the number of people who received their passports in cities A and B be $7m$ and $12m$ respectively.

So, the number of people who did not receive their passports in city A = $(7m - 240)$

And, the number of people who did not receive their passports in city B = $(12m - 360)$

So, the total number of people whose applications were accepted in city A = $7m + (7m - 240) = (14m - 240)$

And, the total number of people whose applications were accepted in city B = $12m + (12m - 360) = (24m - 360)$

So, the number of people whose applications were rejected in city A = $400 - (14m - 240) = (640 - 14m)$

And, the number of people whose applications were rejected in city B = $800 - (24m - 360) = (1160 - 24m)$

Now,

$$(1160 - 24m) - (640 - 14m) = 120$$

$$520 - 10m = 120$$

$$10m = 400$$

$$m = 40$$

City D and E

Let the number of people who received their passport in city D be n .

So, the number of people who received their passport in city E = $(875 - n)$

The number of people who did not receive their passport in city D = $(n - 200)$

And, the number of people who did not receive their passport in city E = $(875 - n) - 300 = (575 - n)$

So, the number of people whose applications were accepted in city D = $n + (n - 200) = (2n - 200)$

And, the number of people whose applications were accepted in city E = $(875 - n) + (575 - n) = (1450 - 2n)$

Now, the number of people whose applications were rejected in city D = $500 - (2n - 200) = (700 - 2n)$

And, the number of people whose applications were rejected in city E = $900 - (1450 - 2n) = (2n - 550)$

So,

$$(700 - 2n) / (2n - 550) = 50/100$$

$$(700 - 2n) / (2n - 550) = 1/2$$

$$1400 - 4n = 2n - 550$$

$$6n = 1950$$

$$n = 325$$

City A

The total number of people who applied = 400

The number of people whose application were accepted = $(14m - 240) = (14 * 40 - 240) = (560 - 240) = 320$

The number of people whose application were rejected = $(400 - 320) = 80$

The number of people who received their passports = $7m = (7 * 40) = 280$

The number of people who did not receive their passports = $(280 - 240) = 40$

City B

The total number of people who applied = 800

The number of people whose applications were accepted = $(24m - 360) = (24 * 40 - 360) = (960 - 360) = 600$

The number of people whose applications were rejected = $(800 - 600) = 200$

The number of people who received their passport = $12m = (12 * 40) = 480$

The number of people who did not receive their passport = $(480 - 360) = 120$

City D

The total number of people who applied = 500

The number of people whose applications were accepted = $(2n - 200) = (2 * 325 - 200) = (650 - 200) = 450$

The number of people whose applications were rejected = $(500 - 450) = 50$

The number of people who received their passport = $n = 325$

The number of people who did not receive their passport = $(325 - 200) = 125$

City E

The total number of people who applied = 900

The number of people whose applications were accepted = $(1450 - 2n) = (1450 - 2 * 325) = (1450 - 650) = 800$

The number of people whose applications were rejected = $(900 - 800) = 100$

The number of people who receive their passport = $(875 - n) = (875 - 325) = 550$

The number of people who did not receive their passport = $(550 - 300) = 250$

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City C

The total number of people who applied = 600

Let the number of people whose applications were rejected be k .

So, the number of people whose applications were accepted = $(600 - k)$

The number of people who did not receive their passport = $(k - 110)$

The number of people who received their passport = $(k - 110) + 280 = (k + 170)$

Now,

$$(k - 110) + (k + 170) = (600 - k)$$

$$2k + 60 = 600 - k$$

$$3k = 540$$

$$k = 180$$

Therefore,

The number of people whose applications were rejected = $k = 180$

The number of people whose applications were accepted = $(600 - k) = (600 - 180) = 420$

The number of people who did not receive their passport = $(k - 110) = (180 - 110) = 70$

The number of people who received their passport = $(k + 170) = (180 + 170) = 350$

City	The total number of people who applied	The number of people whose application were accepted	The number of people whose application were rejected	The number of people who received their passport	The number of people who did not receive their passport
A	400	320	80	280	40
B	800	600	200	480	120
C	600	420	180	350	70
D	500	450	50	325	125
E	900	800	100	550	250
Total	3200	2590	610	1985	605

Answer: A

City C

The total number of people who applied for the driving license = 500

The number of people who received driving license = $(350 - 50) = 300$

Let the number of people who did not receive driving license be x .

So, the number of people whose applications were rejected for driving license = x

Now, the total number people whose applications were accepted for driving license = $(300 + x)$

So,

$$(300 + x) + x = 500$$

$$2x = 200$$

$$x = 100$$

therefore, the total number people whose applications were accepted for driving license in city C = $(300 + x) = (300 + 100) = 400$.

Hence, the correct answer is option A.

13. Questions

City A and B

Let the number of people who received their passports in cities A and B be $7m$ and $12m$ respectively.

So, the number of people who did not receive their passports in city A = $(7m - 240)$

And, the number of people who did not receive their passports in city B = $(12m - 360)$

So, the total number of people whose applications were accepted in city A = $7m + (7m - 240) = (14m - 240)$

And, the total number of people whose applications were accepted in city B = $12m + (12m - 360) = (24m - 360)$

So, the number of people whose applications were rejected in city A = $400 - (14m - 240) = (640 - 14m)$

And, the number of people whose applications were rejected in city B = $800 - (24m - 360) = (1160 - 24m)$

Now,

$$(1160 - 24m) - (640 - 14m) = 120$$

$$520 - 10m = 120$$

$$10m = 400$$

$$m = 40$$

City D and E

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Let the number of people who received their passport in city D be n .

So, the number of people who received their passport in city E = $(875 - n)$

The number of people who did not receive their passport in city D = $(n - 200)$

And, the number of people who did not receive their passport in city E = $(875 - n) - 300 = (575 - n)$

So, the number of people whose applications were accepted in city D = $n + (n - 200) = (2n - 200)$

And, the number of people whose applications were accepted in city E = $(875 - n) + (575 - n) = (1450 - 2n)$

Now, the number of people whose applications were rejected in city D = $500 - (2n - 200) = (700 - 2n)$

And, the number of people whose applications were rejected in city E = $900 - (1450 - 2n) = (2n - 550)$

So,

$$(700 - 2n) / (2n - 550) = 50/100$$

$$(700 - 2n) / (2n - 550) = 1/2$$

$$1400 - 4n = 2n - 550$$

$$6n = 1950$$

$$n = 325$$

City A

The total number of people who applied = 400

The number of people whose application were accepted = $(14m - 240) = (14 * 40 - 240) = (560 - 240) = 320$

The number of people whose application were rejected = $(400 - 320) = 80$

The number of people who received their passports = $7m = (7 * 40) = 280$

The number of people who did not receive their passports = $(280 - 240) = 40$

City B

The total number of people who applied = 800

The number of people whose applications were accepted = $(24m - 360) = (24 * 40 - 360) = (960 - 360) = 600$

The number of people whose applications were rejected = $(800 - 600) = 200$

The number of people who received their passport = $12m = (12 * 40) = 480$

The number of people who did not receive their passport = $(480 - 360) = 120$

City D

The total number of people who applied = 500

The number of people whose applications were accepted = $(2n - 200) = (2 * 325 - 200) = (650 - 200) = 450$

The number of people whose applications were rejected = $(500 - 450) = 50$

The number of people who received their passport = $n = 325$

The number of people who did not receive their passport = $(325 - 200) = 125$

City E

The total number of people who applied = 900

The number of people whose applications were accepted = $(1450 - 2n) = (1450 - 2 * 325) = (1450 - 650)$

= 800

The number of people whose applications were rejected = $(900 - 800) = 100$

The number of people who receive their passport = $(875 - n) = (875 - 325) = 550$

The number of people who did not receive their passport = $(550 - 300) = 250$

City C

The total number of people who applied = 600

Let the number of people whose applications were rejected be k .

So, the number of people whose applications were accepted = $(600 - k)$

The number of people who did not receive their passport = $(k - 110)$

The number of people who received their passport = $(k - 110) + 280 = (k + 170)$

Now,

$$(k - 110) + (k + 170) = (600 - k)$$

$$2k + 60 = 600 - k$$

$$3k = 540$$

$$k = 180$$

Therefore,

The number of people whose applications were rejected = $k = 180$

The number of people whose applications were accepted = $(600 - k) = (600 - 180) = 420$

The number of people who did not receive their passport = $(k - 110) = (180 - 110) = 70$

The number of people who received their passport = $(k + 170) = (180 + 170) = 350$

City	The total number of people who applied	The number of people whose application were accepted	The number of people whose application were rejected	The number of people who received their passport	The number of people who did not receive their passport
A	400	320	80	280	40
B	800	600	200	480	120
C	600	420	180	350	70
D	500	450	50	325	125
E	900	800	100	550	250
Total	3200	2590	610	1985	605

Answer: D

The number of people whose application for passport got rejected in city,

$$A = 80$$

$$B = 200$$

$$C = 180$$

$$D = 50$$

$$E = 100$$

Therefore, the required average = $(80 + 200 + 180 + 50 + 100)/5 = (610/5) = 122$.

Hence, the correct answer is option D.

14. Questions

City A and B

Let the number of people who received their passports in cities A and B be $7m$ and $12m$ respectively.

So, the number of people who did not receive their passports in city A = $(7m - 240)$

And, the number of people who did not receive their passports in city B = $(12m - 360)$

So, the total number of people whose applications were accepted in city A = $7m + (7m - 240) = (14m - 240)$

And, the total number of people whose applications were accepted in city B = $12m + (12m - 360) = (24m - 360)$

So, the number of people whose applications were rejected in city A = $400 - (14m - 240) = (640 - 14m)$

And, the number of people whose applications were rejected in city B = $800 - (24m - 360) = (1160 - 24m)$

Now,

$$(1160 - 24m) - (640 - 14m) = 120$$

$$520 - 10m = 120$$

$$10m = 400$$

$$m = 40$$

City D and E

Let the number of people who received their passport in city D be n .

So, the number of people who received their passport in city E = $(875 - n)$

The number of people who did not receive their passport in city D = $(n - 200)$

And, the number of people who did not receive their passport in city E = $(875 - n) - 300 = (575 - n)$

So, the number of people whose applications were accepted in city D = $n + (n - 200) = (2n - 200)$

And, the number of people whose applications were accepted in city E = $(875 - n) + (575 - n) = (1450 - 2n)$

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Now, the number of people whose applications were rejected in city D = $500 - (2n - 200) = (700 - 2n)$

And, the number of people whose applications were rejected in city E = $900 - (1450 - 2n) = (2n - 550)$

So,

$$(700 - 2n) / (2n - 550) = 50/100$$

$$(700 - 2n) / (2n - 550) = 1/2$$

$$1400 - 4n = 2n - 550$$

$$6n = 1950$$

$$n = 325$$

City A

The total number of people who applied = 400

The number of people whose application were accepted = $(14m - 240) = (14 * 40 - 240) = (560 - 240) = 320$

The number of people whose application were rejected = $(400 - 320) = 80$

The number of people who received their passports = $7m = (7 * 40) = 280$

The number of people who did not receive their passports = $(280 - 240) = 40$

City B

The total number of people who applied = 800

The number of people whose applications were accepted = $(24m - 360) = (24 * 40 - 360) = (960 - 360) = 600$

The number of people whose applications were rejected = $(800 - 600) = 200$

The number of people who received their passport = $12m = (12 * 40) = 480$

The number of people who did not receive their passport = $(480 - 360) = 120$

City D

The total number of people who applied = 500

The number of people whose applications were accepted = $(2n - 200) = (2 * 325 - 200) = (650 - 200) = 450$

The number of people whose applications were rejected = $(500 - 450) = 50$

The number of people who received their passport = $n = 325$

The number of people who did not receive their passport = $(325 - 200) = 125$

City E

The total number of people who applied = 900

The number of people whose applications were accepted = $(1450 - 2n) = (1450 - 2 * 325) = (1450 - 650)$

= 800

The number of people whose applications were rejected = $(900 - 800) = 100$

The number of people who receive their passport = $(875 - n) = (875 - 325) = 550$

The number of people who did not receive their passport = $(550 - 300) = 250$

City C

The total number of people who applied = 600

Let the number of people whose applications were rejected be k .

So, the number of people whose applications were accepted = $(600 - k)$

The number of people who did not receive their passport = $(k - 110)$

The number of people who received their passport = $(k - 110) + 280 = (k + 170)$

Now,

$$(k - 110) + (k + 170) = (600 - k)$$

$$2k + 60 = 600 - k$$

$$3k = 540$$

$$k = 180$$

Therefore,

The number of people whose applications were rejected = $k = 180$

The number of people whose applications were accepted = $(600 - k) = (600 - 180) = 420$

The number of people who did not receive their passport = $(k - 110) = (180 - 110) = 70$

The number of people who received their passport = $(k + 170) = (180 + 170) = 350$

City	The total number of people who applied	The number of people whose application were accepted	The number of people whose application were rejected	The number of people who received their passport	The number of people who did not receive their passport
A	400	320	80	280	40
B	800	600	200	480	120
C	600	420	180	350	70
D	500	450	50	325	125
E	900	800	100	550	250
Total	3200	2590	610	1985	605

Answer: C

City F

The total number of people who applied = 60% of 900 = 540

The number of people whose applications were rejected = $\frac{1}{9}$ of 540 = 60

Therefore, the number of people whose applications were accepted in city F = $(540 - 60) = 480$

Hence, the correct answer is option C.

15. Questions

City A and B

Let the number of people who received their passports in cities A and B be $7m$ and $12m$ respectively.

So, the number of people who did not receive their passports in city A = $(7m - 240)$

And, the number of people who did not receive their passports in city B = $(12m - 360)$

So, the total number of people whose applications were accepted in city A = $7m + (7m - 240) = (14m - 240)$

And, the total number of people whose applications were accepted in city B = $12m + (12m - 360) = (24m - 360)$

So, the number of people whose applications were rejected in city A = $400 - (14m - 240) = (640 - 14m)$

And, the number of people whose applications were rejected in city B = $800 - (24m - 360) = (1160 - 24m)$

Now,

$$(1160 - 24m) - (640 - 14m) = 120$$

$$520 - 10m = 120$$

$$10m = 400$$

$$m = 40$$

City D and E

Let the number of people who received their passport in city D be n .

So, the number of people who received their passport in city E = $(875 - n)$

The number of people who did not receive their passport in city D = $(n - 200)$

And, the number of people who did not receive their passport in city E = $(875 - n) - 300 = (575 - n)$

So, the number of people whose applications were accepted in city D = $n + (n - 200) = (2n - 200)$

And, the number of people whose applications were accepted in city E = $(875 - n) + (575 - n) = (1450 - 2n)$

Now, the number of people whose applications were rejected in city D = $500 - (2n - 200) = (700 - 2n)$

And, the number of people whose applications were rejected in city E = $900 - (1450 - 2n) = (2n - 550)$

So,

$$(700 - 2n) / (2n - 550) = 50/100$$

$$(700 - 2n) / (2n - 550) = 1/2$$

$$1400 - 4n = 2n - 550$$

$$6n = 1950$$

$$n = 325$$

City A

The total number of people who applied = 400

The number of people whose application were accepted = $(14m - 240) = (14 * 40 - 240) = (560 - 240) = 320$

The number of people whose application were rejected = $(400 - 320) = 80$

The number of people who received their passports = $7m = (7 * 40) = 280$

The number of people who did not receive their passports = $(280 - 240) = 40$

City B

The total number of people who applied = 800

The number of people whose applications were accepted = $(24m - 360) = (24 * 40 - 360) = (960 - 360) = 600$

The number of people whose applications were rejected = $(800 - 600) = 200$

The number of people who received their passport = $12m = (12 * 40) = 480$

The number of people who did not receive their passport = $(480 - 360) = 120$

City D

The total number of people who applied = 500

The number of people whose applications were accepted = $(2n - 200) = (2 * 325 - 200) = (650 - 200) = 450$

The number of people whose applications were rejected = $(500 - 450) = 50$

The number of people who received their passport = $n = 325$

The number of people who did not receive their passport = $(325 - 200) = 125$

City E

The total number of people who applied = 900

The number of people whose applications were accepted = $(1450 - 2n) = (1450 - 2 * 325) = (1450 - 650) = 800$

The number of people whose applications were rejected = $(900 - 800) = 100$

The number of people who receive their passport = $(875 - n) = (875 - 325) = 550$

The number of people who did not receive their passport = $(550 - 300) = 250$

City C

The total number of people who applied = 600

Let the number of people whose applications were rejected be k .

So, the number of people whose applications were accepted = $(600 - k)$

The number of people who did not receive their passport = $(k - 110)$

The number of people who received their passport = $(k - 110) + 280 = (k + 170)$

Now,

$$(k - 110) + (k + 170) = (600 - k)$$

$$2k + 60 = 600 - k$$

$$3k = 540$$

$$k = 180$$

Therefore,

The number of people whose applications were rejected = $k = 180$

The number of people whose applications were accepted = $(600 - k) = (600 - 180) = 420$

The number of people who did not receive their passport = $(k - 110) = (180 - 110) = 70$

The number of people who received their passport = $(k + 170) = (180 + 170) = 350$

City	The total number of people who applied	The number of people whose application were accepted	The number of people whose application were rejected	The number of people who received their passport	The number of people who did not receive their passport
A	400	320	80	280	40
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D	500	450	50	325	125
E	900	800	100	550	250
Total	3200	2590	610	1985	605

Answer: B

The number of people who received their passport in city A = 280

The number of people who received their passport in city D = 325

Therefore, the required ratio = $280:325 = 56:65$.

Hence, the correct answer is option B.

16. Questions

Answer: A

Series I: 10, 14, -2, [A], -30, [B]

$$1^{\text{st}} \text{ term} = 10$$

$$2^{\text{nd}} \text{ term} = (10 + 2^2) = (10 + 4) = 14$$

$$3^{\text{rd}} \text{ term} = (14 - 4^2) = (14 - 16) = -2$$

$$4^{\text{th}} \text{ term} = (-2 + 6^2) = (-2 + 36) = 34 \text{ (A)}$$

$$5^{\text{th}} \text{ term} = (34 - 8^2) = (34 - 64) = -30$$

$$6^{\text{th}} \text{ term} = (-30 + 10^2) = (-30 + 100) = 70 \text{ (B)}$$

So, the missing terms = 34 and 70

$$A = 34$$

$$B = 70$$

Series II: 15, [P], 3, [Q], -25, 75

$$1^{\text{st}} \text{ term} = 15$$

$$2^{\text{nd}} \text{ term} = (15 + 2^2) = (15 + 4) = 19 \text{ (P)}$$

$$3^{\text{rd}} \text{ term} = (19 - 4^2) = (19 - 16) = 3$$

$$4^{\text{th}} \text{ term} = (3 + 6^2) = (3 + 36) = 39 \text{ (Q)}$$

$$5^{\text{th}} \text{ term} = (39 - 8^2) = (39 - 64) = -25$$

$$6^{\text{th}} \text{ term} = (-25 + 10^2) = (-25 + 100) = 75$$

$$P = 19$$

$$Q = 39$$

I: 39 is not a negative integer.

So, I is not true.

$$\text{II: } (A - P) = (34 - 19) = 15$$

So, II is not true.

$$\text{III: } (B * Q) = (70 * 39) = 2730$$

2730 is four-digit natural number.

So, III is true.

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Therefore, only III must be true.

Hence, the correct answer is option A.

17. Questions

Answer: E

Series I: 24, 48, 12, 30, 6, 12

$$1^{\text{st}} \text{ term} = 24$$

$$2^{\text{nd}} \text{ term} = (24 * 2) = 48$$

$$3^{\text{rd}} \text{ term} = (48 / 4) = 12$$

$$4^{\text{th}} \text{ term} = (12 * 2) = 24 \text{ (30 is wrong)}$$

$$5^{\text{th}} \text{ term} = (24 / 4) = 6$$

$$6^{\text{th}} \text{ term} = (6 * 2) = 12$$

So, the wrong term = 30

And, the correct term = 24

Series II: ____, 51, 115, 240, 456, __[?]

24, 51, 115, 240, 456, __[?]

$$1^{\text{st}} \text{ term} = 24$$

$$2^{\text{nd}} \text{ term} = (24 + 3^3) = (24 + 27) = 51$$

$$3^{\text{rd}} \text{ term} = (51 + 4^3) = (51 + 64) = 115$$

$$4^{\text{th}} \text{ term} = (115 + 5^3) = (115 + 125) = 240$$

$$5^{\text{th}} \text{ term} = (240 + 6^3) = (240 + 216) = 456$$

$$6^{\text{th}} \text{ term} = (456 + 7^3) = (456 + 343) = 799 \text{ (?)}$$

So, the missing term = ? = 799

Hence, the correct answer is option E.

18. Questions

Answer: A

Series I: 12, 23, 36, 53, 70, 95

$$1^{\text{st}} \text{ term} = 12$$

$$2^{\text{nd}} \text{ term} = (12 + 11) = 23$$

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$$3^{\text{rd}} \text{ term} = (23 + 13) = 36$$

$$4^{\text{th}} \text{ term} = (36 + 17) = 53$$

$$5^{\text{th}} \text{ term} = (53 + 19) = 72 \text{ (70 is wrong)}$$

$$6^{\text{th}} \text{ term} = (72 + 23) = 95$$

So, the wrong term = 70

And, the correct term = 72

Series II: 31, 42, 55, 72, 91, 110

$$1^{\text{st}} \text{ term} = 31$$

$$2^{\text{nd}} \text{ term} = (31 + 11) = 42$$

$$3^{\text{rd}} \text{ term} = (42 + 13) = 55$$

$$4^{\text{th}} \text{ term} = (55 + 17) = 72$$

$$5^{\text{th}} \text{ term} = (72 + 19) = 91$$

$$6^{\text{th}} \text{ term} = (91 + 23) = 114 \text{ (110 is wrong)}$$

So, the wrong term = 110

And, the correct term = 114

Now,

$$\sqrt{M} + 2 = (7 - 2)$$

$$[\sqrt{M} + 2 = 5]$$

$$\sqrt{M} = 3$$

$$M = 3^2$$

$$M = 9$$

Also,

$$N^2 = 4^2$$

$$N^2 = 16$$

$$N = \sqrt{16}$$

$$N = 4$$

Therefore, $M > N$.

Hence, the correct answer is option A.

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19. Questions

Answer: E

Series I: 8, 21, 58, 243, __[X]__, 7265

$$1^{\text{st}} \text{ term} = 8$$

$$2^{\text{nd}} \text{ term} = (8 * 2 + 5) = (16 + 5) = 21$$

$$3^{\text{rd}} \text{ term} = (21 * 3 - 5) = (63 - 5) = 58$$

$$4^{\text{th}} \text{ term} = (58 * 4 + 5) = (232 + 5) = 237$$

$$5^{\text{th}} \text{ term} = (237 * 5 - 5) = (1185 - 5) = 1180 (X)$$

$$6^{\text{th}} \text{ term} = (1180 * 6 + 5) = (7080 + 5) = 7085$$

So, the missing term = $X = 1180$

Series II: $[X+30/110]$, 19, 35, 59, 91, __[Y]__

$[1210/110]$, 19, 35, 59, 91, __[Y]__

$$1^{\text{st}} \text{ term} = 11$$

$$2^{\text{nd}} \text{ term} = (11 + 8) = 19$$

$$3^{\text{rd}} \text{ term} = (19 + 16) = 35$$

$$4^{\text{th}} \text{ term} = (35 + 24) = 59$$

$$5^{\text{th}} \text{ term} = (59 + 32) = 91$$

$$6^{\text{th}} \text{ term} = (91 + 40) = 131 (Y)$$

So, the missing term = $Y = 131$

Therefore, the value of $= (10Y - X) = (10 * 131 - 1180) = (1310 - 1180) = 130$.

Hence, the correct answer is option E.

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20. Questions

Answer: B

Series I: 10, 22, 69, 280, __[m]__, 8436

$$1^{\text{st}} \text{ term} = 10$$

$$2^{\text{nd}} \text{ term} = (10 * 2 + 2) = (20 + 2) = 22$$

$$3^{\text{rd}} \text{ term} = (22 * 3 + 3) = (66 + 3) = 69$$

$$4^{\text{th}} \text{ term} = (69 * 4 + 4) = (276 + 4) = 280$$

$$5^{\text{th}} \text{ term} = (280 * 5 + 5) = (1400 + 5) = 1405 \text{ (m)}$$

$$6^{\text{th}} \text{ term} = (1405 * 6 + 6) = (8430 + 6) = 8436$$

So, the missing term = m = 1405

Series II: [(m - 5)/280], 12, 39, 156, 805, 4836

$$[(1405 - 5)/280], 12, 39, 156, 805, 4836$$

$$[1400/280], 12, 39, 156, 805, 4836$$

$$5, 12, 39, 156, 805, 4836$$

$$1^{\text{st}} \text{ term} = 5$$

$$2^{\text{nd}} \text{ term} = (5 * 2 + 2) = (10 + 2) = 12$$

$$3^{\text{rd}} \text{ term} = (12 * 3 + 3) = (36 + 3) = 39$$

$$4^{\text{th}} \text{ term} = (39 * 4 + 4) = (156 + 4) = 160 \text{ (156 is wrong)}$$

$$5^{\text{th}} \text{ term} = (160 * 5 + 5) = (800 + 5) = 805$$

$$6^{\text{th}} \text{ term} = (805 * 6 + 6) = (4830 + 6) = 4836$$

So, the wrong term = 156

And, the correct term = 160

Hence, the correct answer is option B.

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